Chapter 1: Algebraic expressions

Chapter 1 · Original slide 1 · Algebraic expressions

Pure Y1 Ch 1
Algebraic Expressions

Chapter 1 · Original slide 2 · Algebraic Expression Starter

1) Simplify:

b) $4m^2n + 5mn^2 - 2m^2n + mn^2 - 3mn^2$

c) $3x^2 - 5x + 2 + 3x^2 - 7x - 12$

5) Write as a single power of 2:

f) $2^5 \times 2^3$ b) $2^6 \div 2^2$ c) $\left(2^3\right)^2$

8) Expand:

i) $3\left(x + 4\right)$ b) $5\left(2 - 3x\right)$ c) $6\left(2x - 5y\right)$

11) Write down the highest common factor of:

l) 24 and 16

m) $6x$ and $8x^2$

n) $4xy^2$ and $3xy$

16) Simplify:

q) $\frac{10x}{5}$ b) $\frac{20x}{2}$ c) $\frac{40x}{24}$

Chapter 1: Algebraic expressions

Chapter 1 · Original slide 3 · Chapter Overview

As a relatively gentle introduction to Pure, most of this chapter is a recap of core GCSE algebraic skills.

1: Basic Index Laws

2: Expand brackets

Simplify
$\frac{3x^2-6x^5}{2x}$

Expand and simplify
$(x-3)^2(x+1)$

4: Fractional/Negative Powers

Factorise fully:
x^3-16x

3: Factorise quadratics/cubics

Evaluate:
$\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

Rationalise the denominator of
$\frac{1}{3+\sqrt{2}}$

5: Surds

Chapter 1 · Original slide 4 · 1: Index Laws

base

3^5

exponent or index (plural: indices)

!

$a^m \times a^n = a^{m+n}$
$a^m \div a^n = a^{m-n}$
$(a^m)^n = a^{mn}$
$(ab)^n = a^n b^n$

power

Some also refer to the 5 as the ‘power’. But this would lead to unfortunate ambiguous phrases like “when we multiply two powers together, we add the powers” instead of “when we multiply two powers together, we add the indices”.

Simplify $\left(a^3\right)^2 \times 2a^2$

Simplify $\left(4x^3y\right)^3$

Simplify $2x^2(3+5x) - x\left(4-x^2\right)$

Simplify $\frac{x^3-2x}{3x^2}$

Tip: A common student error is to get the sign wrong of $+x^3$

Tip: While $\frac{a+b}{c}$ can be split into $\frac{a}{c}+\frac{b}{c}$, a common student error is to think that $\frac{a}{b+c} = \frac{a}{b}+\frac{a}{c}$

Chapter 1: Algebraic expressions

Chapter 1 · Original slide 5 · Test Your Understanding

1

Simplify $\left(\frac{2a^5}{a^2}\right)^2 \times 3a$

2

Simplify $\frac{2x+x^5}{4x^3}$

3

Expand and simplify

$2x\left(3-x^2\right)-4x^3\left(3-x\right)$

4

Simplify $2^x \times 3^x$

Note: This is using $(ab)^n = a^n b^n$ law backwards.

Chapter 1 · Original slide 6 · Exercise 1A – Pearson Pure Y1 page 4

2 Expand and simplify if possible:

o $x(3x^2-2x+5)$
p $7y^2(2-5y+3y^2)$
q $-2y^2(5-7y+3y^2)$
r $7(x-2)+3(x+4)-6(x-2)$
s $5x-3(4-2x)+6$
t $3x^2-x(3-4x)+7$
u $4x(x+3)-2x(3x-7)$
v $3x^2(2x+1)-5x^2(3x-4)$

d $\frac{8x^3+5x}{2x}$ **e** $\frac{7x^7+5x^2}{5x}$ **f** $\frac{9x^5-5x^3}{3x}$

3 Simplify these fractions:

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Chapter 1 · Original slide 7 · Exercise 1A – Pearson Pure Y1 page 4

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1

[MAT 2006 1A] Which of the following numbers is largest?

$$\left((2^3)^2\right)^3, \quad (2^3)^{(2^3)}, \quad 2^{((3^2)^3)}, \quad 2^{(3^{(2^3)})}$$

2

[MAT 2012 1B] Let $N = 2^k \times 4^m \times 8^n$ where k,m,n are positive whole numbers. Then N will definitely be a square number whenever:

k is even;

$k + n$ is odd;

k is odd but $m + n$ is even;

$k + n$ is even.

Chapter 1 · Original slide 8 · 2: Expanding Brackets

If you have ever been taught 'FOIL' to multiply brackets please purge it from your mind now – instead:

Multiply each term in the first bracket by each term in the second.

$$(x - y)(x + y - 1)$$

Tip: My order is "first term in first brackets times each in second, then second term in first bracket times each in second, etc."

Chapter 1: Algebraic expressions

Chapter 1 · Original slide 9 · Test Your Understanding

1

Expand and simplify

$(x+5)(x-2)(x+1)$

2

Expand and simplify:

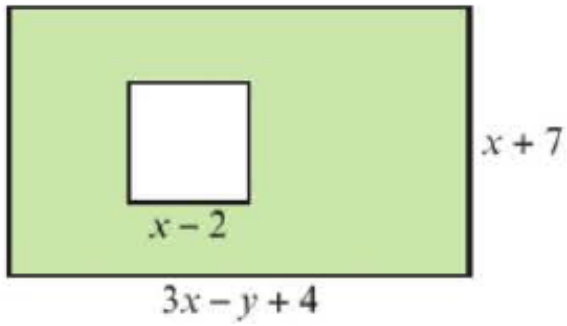
$2(x-3)(x-4)$

3

Expand and simplify:

$(2x-1)^3$

Chapter 1 · Original slide 10 · Exercise 1B - Pure Year 1/AS Page 6



3 The diagram shows a rectangle with a square cut out. The rectangle has length 3x-y+4 and width x+7. The square has length x-2. Find an expanded and simplified expression for the shaded area.

5 Given that
$(2x+5y)(3x-y)(2x+y) = ax^3 + bx^2y + cxy^2 + dy^3$,
where a,b,c,d are constants, find the values of a,b,c,d.

Chapter 1: Algebraic expressions

Chapter 1 · Original slide 11 · Exercise 1B - Pure Year 1/AS Page 5

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1

[MAT 2002 1B] Of the following three alleged algebraic identities, at least one is wrong.

(i) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(x-z)(y-x)$

(ii) $yz(z-y)+zx(x-z)+xy(y-x)$
$=(z-y)(z-x)(y-x)$

(iii) $yz(x+y)+zx(z+x)+xy(y+x)$
$=(z+y)(z+x)(y+x)$

Which of the following statements are correct? Tick all that apply. (i), (ii), (iii).

2

[MAT 2007 1E] If x and n are integers then $(1-x)^n(2-x)^{2n}(3-x)^{3n}(4-x)^{4n}(5-x)^{5n}$ is:

negative when $n>5$ and $x<5$

negative when n is odd and $x>5$

negative when n is a multiple of 3 and $x>5$

negative when n is even and $x<5$

Chapter 1 · Original slide 12 · 3: Factorising

Informally, factorising is the opposite of expanding brackets.

More formally, a factorised expression is one which is expressed as a product of expressions.

$x(x+1)(x+2)$

Factorised as it is the product of 3 linear factors, $x, x+1$ and $x+2$.

Note: A linear expression is of the form $ax+b$. It is called linear because plotting $y=ax+b$ would form a straight line.

Not factorised because the outer-most operation is a sum, not a product.

$x\left(x+1\right)+\left(x-1\right)\left(x+1\right)$

Chapter 1 · Original slide 13 · Factorising Quadratics

Recap:

$\oplus$

$x^2 - 5x - 14 =$

We find two numbers which multiply to give the coefficient of x and multiply to give the constant term.

$\otimes$

Note: The *coefficient* of a term is the constant on front of it, e.g. the coefficient of $4x^2$ is 4.

Chapter 1 · Original slide 14 · Other Factorisations

Difference of two squares:

$4x^2 - 9 =$

Using multiple factorisations:

$x^3 - x$

Tip: Always look for a common factor first before using other factorisation techniques.

$x^3 + 3x^2 + 2x$

Chapter 1 · Original slide 15 · Test Your Understanding

2

Factorise completely:

$x^3 - 7x^2 + 12x$

N

Factorise completely:

$x^4 - 1$

Note: You would not be expected to factorise this at A Level (but you would in STEP!).

In general, the *difference of two cubes*:

$x^3 - y^3 = (x - y)\left(x^2 + xy + y^2\right)$

1

Factorise completely:

$6x^2 + x - 2$

N

Factorise completely:

$x^3 - 1$

Chapter 1 · Original slide 16 · Exercise 1C - Pure Y1 Page 8

4 Factorise completely $x^4 - y^4$.

Challenge: Write $4x^4 - 13x^2 + 9$as the product of four linear factors

5 Factorise completely $6x^3 + 7x^2 - 5x$.

Chapter 1: Algebraic expressions

Chapter 1 · Original slide 17 · 4: Negative and Fractional Indices

$a^0 = 1$
$a^{\frac{1}{m}} = \sqrt[m]{a}$
$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$
$a^{-m} = \frac{1}{a^m}$

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $32^{\frac{2}{5}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.

Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Evaluate $27^{-\frac{1}{3}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

Chapter 1 · Original slide 18 · Writing a surd using indices

If $9\sqrt{3} = 3^k$, determine the value of k.

Note: In algebra we like to avoid mixed numbers. So don't write $3^{2\frac{1}{2}}$

The key here is to write everything as powers with a consistent base, in this case, 3.

Chapter 1 · Original slide 19 · Further Examples

[Edexcel IGCSE May14-4H] Given that $\left(2^{\frac{1}{2}}\right)^n = \frac{2^x}{8^y}$

Express n in terms of x and y.

Write 8 as a power of 2 (putting a bracket around it), for consistency of base with the other powers.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^x$, express the following in terms of y.

$\frac{1}{4^{2x-3}}$

Write your expression in its simplest form.

Chapter 1 · Original slide 20 · Test Your Understanding

Given $2^x \times 4^y = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

Chapter 1 · Original slide 21 · Exercise 1D

4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants.
a $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

Chapter 1 · Original slide 22 · 5: Surds

Recap:

Note: A *rational* number is any which can be expressed as $\frac{a}{b}$ where a, b are integers. $\frac{2}{3}$ and $\frac{4}{1} = 4$ are rational numbers, but π and $\sqrt{2}$ are not.

A surd is a root of a number that does not simplify to a rational number.

Laws:

$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$

$\sqrt{3} \times 2$
$3\sqrt{5} \times 2\sqrt{5}$
$\sqrt{8} = \sqrt{4}\sqrt{2}$
$\sqrt{12} + \sqrt{27}$
$\left(\sqrt{8} + 1\right)\left(\sqrt{2} - 3\right)$

Chapter 1 · Original slide 23 · Exercise 1E – Pure Y1 Page 13

3 Simplify $\sqrt{75} - \sqrt{12}$ giving your answer in the form $a\sqrt{3}$, where a is an integer.

Extension

1

[SMC 2014 Q24] Which of the following is smallest?

$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$; $9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$.

2

[SMC 2012 Q21] Which of the following numbers does not have a square root in the form $x + y\sqrt{2}$, where x and y are positive integers?

$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$; $54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$.

Chapter 1 · Original slide 24 · 6: Rationalising The Denominator

Here's a surd. What could we multiply it by such that it's no longer an irrational number?

$\sqrt{5}\times$

$\frac{1}{\sqrt{2}}$

In this fraction, the denominator is irrational. 'Rationalising the denominator' means making the denominator a rational number.

What could we multiply this fraction by to both rationalise the denominator, but leave the value of the fraction unchanged?

Side Note: There's two reasons why we might want to do this:

1. For aesthetic reasons, it makes more sense to say "half of root 2" rather than "one root two-th of 1". It's nice to divide by something whole!

2. It makes it easier for us to add expressions involving surds.

Chapter 1 · Original slide 25 · Examples

$\frac{3}{\sqrt{2}} =$

$\frac{6}{\sqrt{3}} =$

$\frac{7}{\sqrt{7}} =$

$\frac{15}{\sqrt{5}} + \sqrt{5} =$

Chapter 1 · Original slide 26 · More Complex Denominators

You've seen 'rationalising a denominator', the idea being that we don't like to divide things by an irrational number.

We basically use the same expression but with the sign reversed (this is known as the *conjugate*). That way, we obtain the difference of two squares. Since $(a+b)(a-b) = a^2 - b^2$, any surds will be squared and thus we'll end up with no surds in the denominator.

But what do we multiply the top and bottom by if we have a more complicated denominator?

$\frac{1}{\sqrt{2}+1} \times$

Chapter 1 · Original slide 27 · More Examples

$\frac{3}{\sqrt{6}-2} \times$

You can explicitly expand out $\left(\sqrt{6}-2\right)\left(\sqrt{6}+2\right)$ in the denominator, but remember that $(a-b)(a+b)=a^2-b^2$ so we can mentally obtain $6-4=2$

Just remember: 'difference of two squares'!

$\frac{4}{\sqrt{3}+1} \times$

$\frac{3\sqrt{2}+4}{5\sqrt{2}-7} \times$

Chapter 1 · Original slide 28 · Test Your Understanding

Rationalise the denominator and simplify

$\frac{4}{\sqrt{5}-2}$

Solve $y\left(\sqrt{3}-1\right)=8$

Give your answer in the form $a+b\sqrt{3}$ where a and b are integers.

Rationalise the denominator and simplify

$\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$

Chapter 1 · Original slide 29 · Writing surd expressions in power form

This is not in the textbook, but a common type of question in exams, often asked in conjunction with differentiation (a later chapter).

Express $\frac{(2+\sqrt{x})^2}{x}$ as powers of x.

Write any roots as powers of x.

Test Your Understanding

Express $\frac{(x+\sqrt{x})^2}{2\sqrt{x}}$ as powers of x.

Expand.

Split fraction (some may wish to do this step mentally)

Chapter 1 · Original slide 30 · Exercise 1F (Page 15)

or alternatively: (not in textbook)

2 3 4 5 6

Expand and simplify:

$\left(\sqrt{5}+3\right)\left(\sqrt{5}-2\right)\left(\sqrt{5}+1\right) =$

Rationalise the denominator, giving your answer in the form a+b√3.

$\frac{3\sqrt{3}+7}{3\sqrt{3}-5} =$

Solve x(4−√6)=10 giving your answer in the form a+b√6.

Solve y(1+√2)−√2=3

Simplify:

$\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}} =$

1 a b c d e

Rationalise the denominator and simplify the following:

$\frac{1}{\sqrt{5}+2} =$

$\frac{\sqrt{3}}{\sqrt{3}-1} =$

$\frac{\sqrt{5}+1}{\sqrt{5}-2} =$

$\frac{2\sqrt{3}-1}{3\sqrt{3}+4} =$

$\frac{5\sqrt{5}-2}{2\sqrt{5}-3} =$

Chapter 1: Algebraic expressions

Chapter 1 · Original slide 31 · A final super hard puzzle

N

Solve $\frac{\sqrt[4]{9}}{\sqrt[5]{27}} = \sqrt[x]{3}$